

Software Requirements Specification (SRS)

for

WealthWise – AI-Powered Personal Wealth Planner

Version 1.0 Approved

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1. Introduction

1.1 Purpose

This document specifies the software requirements for WealthWise, an AI-powered personal wealth planning platform. It defines the functional requirements, nonfunctional requirements, external interfaces, and system features needed to design, build, and evaluate the system. The scope covers personal financial data management, wealth analysis, AI-assisted advisory, and forward-looking financial planning tools.

1.2 Document Conventions

Convention	Meaning
FR-xx	Functional Requirement
NFR-xx	Nonfunctional Requirement
BR-xx	Business Rule
OR-xx	Other Requirement
"Shall"	Mandatory requirement

1.3 Intended Audience and Reading Suggestions

Audience	Purpose
Developers	Implementation guidance
Testers	Validation of requirements
Project Managers	Scope tracking
Users	Understanding system capabilities
Faculty Evaluators	Academic grading

1.4 Product Scope

WealthWise is a personal finance and wealth management platform that helps individuals manage their income, expenses, assets, liabilities, investments, savings, and financial goals in one place. The platform analyzes a user's complete financial picture and uses AI to explain what the numbers mean, highlight risks, and support better financial decisions. Beyond basic tracking, WealthWise introduces five innovative planning engines that assess financial health, test resilience against shocks, detect conflicts between competing goals, evaluate the impact of financial decisions, and simulate future scenarios.

Key highlights include:

- Unified Financial View: All income, expenses, assets, liabilities, and investments consolidated in one dashboard.
- AI-Assisted Insight: Plain-language explanations of financial health, risk, and recommended actions.
- Forward-Looking Planning: Scenario simulation and reverse planning to work backward from a goal.

- Resilience Awareness: Stress testing against events such as job loss or medical emergencies.
- Goal-Aware Decisioning: Detection of conflicts between goals and prioritization support.

This scope ensures the platform helps users understand their current financial position and make informed decisions about their financial future, while remaining maintainable, secure, and easy to extend.

1.5 References

Reference	Description
IEEE Std 830-1998	Software Requirements Specification
RFC 8446 (TLS 1.3)	Secure Communication Guidelines
OWASP ASVS	Application Security Verification Standard
UML 2.0	Diagram Specification

2. Overall Description

2.1 Product Perspective

Aspect	Description
Type	Standalone web application with responsive mobile-friendly design
Integration	External AI service (OpenAI or Gemini API) for advisory explanations
Relation	A new, independent product; not affiliated with any bank or financial institution

2.2 Product Functions

Function	Description
Account & Data Management	Signup, login, and management of income, expenses, assets, liabilities, and investments
Wealth Scoring	Calculates a financial health score and net worth summary
Stress Testing	Evaluates resilience against financial shocks
Goal Conflict Detection	Identifies competing goals and recommends priorities
Decision Impact Analysis	Assesses how a proposed decision affects the user's finances
Scenario Simulation	Projects future outcomes and supports reverse planning from a goal
Dashboard & Reporting	Visual summaries, charts, and downloadable reports

2.3 User Classes and Characteristics

User Class	Characteristics
Users	Track personal finances, set goals, and view AI-generated insights
Administrators	Monitor platform usage, manage AI configuration, and enforce policies

2.4 Operating Environment

Component	Environment
Web Client	React.js single-page application; Chrome, Edge, Firefox, Safari
Backend	Node.js with Express.js, hosted on cloud infrastructure
Database	MySQL relational database
AI Layer	OpenAI API or Gemini API accessed over HTTPS

2.5 Design and Implementation Constraints

Constraint	Description
Authentication	JWT-based session management is mandatory
Protocol	HTTPS mandatory for all client-server communication
Data Sensitivity	Financial data must be encrypted at rest and in transit
Charting	Chart.js or Recharts used for all visual dashboards

2.6 User Documentation

Format	Description
Online Help	In-app FAQs and guided tooltips
User Guide (PDF)	Step-by-step guide covering setup and key features

2.7 Assumptions and Dependencies

Assumption	Description
Internet	Users have a stable internet connection
AI Availability	The third-party AI API (OpenAI/Gemini) is available and reachable
Data Accuracy	Users provide reasonably accurate financial information

2.8 SDLC Model

SDLC Model Used: Agile Model

WealthWise will be developed using the **Agile Software Development Life Cycle (SDLC) model**. Agile follows an iterative and incremental approach in which the system is developed in small, manageable increments called iterations or sprints.

The WealthWise project will be divided into multiple development iterations. Each iteration will focus on implementing, testing, and improving a specific set of features. This approach allows the development team to continuously test the system, identify issues, incorporate feedback, and improve the product throughout the development process.

The Agile model is suitable for WealthWise because:

- Requirements and features may evolve during development.
- AI-based features require continuous testing and refinement.
- Individual modules can be developed and tested independently.
- Feedback can be incorporated during development.
- Testing and debugging can be performed continuously.
- New features and improvements can be added without redesigning the entire system.

The major development iterations for WealthWise are:

Iteration 1: Project setup, database design, user registration and authentication.

Iteration 2: Income, expense, asset, liability, and investment management.

Iteration 3: Financial Health and Wealth Scoring Engine.

Iteration 4: Financial Stress Testing and Resilience Analysis.

Iteration 5: Goal Conflict Detection and Priority Engine.

Iteration 6: Financial Decision Impact and Alternative Recommendation Engine.

Iteration 7: Future Financial Scenario and Reverse Planning Engine.

Iteration 8: Dashboard, reports, AI integration, testing, and final deployment.

3. External Interface Requirements

Interface Type	Description
User Interfaces	Responsive React.js web UI with accessible, mobile-friendly layouts and dashboard visualizations
Hardware Interfaces	Standard client devices (desktop, laptop, mobile) with no specialized hardware required
Software Interfaces	Node.js/Express.js backend, MySQL database, OpenAI or Gemini API for AI-generated explanations
Communication Interfaces	HTTPS, RESTful APIs, JWT-secured session tokens

4. System Features (Functional Requirements)

The following seven modules define the functional scope of WealthWise. Modules 4.1 through 4.5 are the Most Significant Points (MSP) of the project — the innovative analytical engines that differentiate WealthWise from a conventional expense tracker. Modules 4.6 and 4.7 are supporting modules that provide the data foundation and presentation layer for those engines.

4.1 Financial Health and Wealth Scoring Engine (MSP Module)

Analyzes a user's complete financial data to produce a single, explainable wealth score along with a net worth summary and trend history.

ID	Requirement Statement	Rationale
FR-01	The system shall calculate the user's net worth from total assets minus total liabilities.	Establishes the baseline figure used across all wealth analysis features.
FR-02	The system shall compute a Financial Health / Wealth Score based on income stability, savings rate, debt level, and liquidity.	Gives users a single, easy-to-understand measure of overall financial health.
FR-03	The system shall generate an AI-based plain-language explanation of the wealth score.	Helps non-expert users understand what drives their score.
FR-04	The system shall track changes in the wealth score over time and display the trend.	Allows users to see whether their financial health is improving or declining.
FR-05	The system shall break down the wealth score into sub-factors (savings, debt, liquidity, growth).	Provides actionable detail rather than a single opaque number.

4.2 Financial Stress Testing and Resilience Analysis (MSP Module)

Simulates adverse financial events to measure how long a user could sustain their lifestyle and identifies the weakest point in their financial resilience.

ID	Requirement Statement	Rationale
FR-06	The system shall allow users to run predefined stress scenarios (e.g., job loss, medical emergency, income drop).	Prepares users for real-world financial shocks.
FR-07	The system shall calculate how many months of expenses the user could sustain under a chosen stress scenario.	Quantifies financial resilience in a concrete, relatable metric.
FR-08	The system shall identify the weakest point in the user's finances during a stress scenario (e.g., low emergency fund).	Directs users toward the most impactful corrective action.
FR-09	The system shall generate AI-based resilience recommendations after each stress test.	Converts raw analysis into practical next steps.

4.3 Goal Conflict Detection and Priority Engine (MSP Module)

Detects when multiple financial goals compete for the same limited funds and recommends an adjusted, prioritized contribution plan.

ID	Requirement Statement	Rationale
FR-10	The system shall detect when two or more active financial goals compete for the same funds within an overlapping time frame.	Prevents users from unknowingly overcommitting their income.
FR-11	The system shall rank conflicting goals using user-defined priority and urgency (target date).	Provides an objective basis for prioritization.
FR-12	The system shall recommend an adjusted contribution plan when goal conflicts are detected.	Offers a practical resolution rather than just flagging the problem.
FR-13	The system shall allow users to accept, modify, or dismiss a conflict-resolution recommendation.	Keeps the user in control of final financial decisions.

4.4 Financial Decision Impact and Alternative Recommendation Engine (MSP Module)

Evaluates the projected impact of a proposed financial decision and generates a comparatively better alternative for the user to consider.

ID	Requirement Statement	Rationale
FR-14	The system shall allow users to input a proposed financial decision (e.g., large purchase, loan, career change).	Enables what-if analysis before a decision is finalized.
FR-15	The system shall project the impact of the proposed decision on net worth, savings rate, and existing goals.	Shows the downstream consequences of a decision in concrete terms.
FR-16	The system shall generate at least one alternative decision path with a comparatively better financial outcome.	Gives users a constructive option instead of only a warning.
FR-17	The system shall present a side-by-side comparison of the original decision and the AI-suggested alternative.	Simplifies decision-making through direct visual comparison.

4.5 Future Financial Scenario and Reverse Planning Engine (MSP Module)

Projects long-term financial outcomes under adjustable assumptions and supports reverse planning from a target goal back to a required monthly contribution.

ID	Requirement Statement	Rationale
FR-18	The system shall simulate future net worth and goal progress over a user-selected time horizon.	Helps users visualize long-term outcomes of their current habits.

ID	Requirement Statement	Rationale
FR-19	The system shall support reverse planning, calculating the monthly contribution required to reach a target amount by a target date.	Turns a goal into an actionable monthly plan.
FR-20	The system shall allow users to adjust simulation assumptions (return rate, inflation, contribution amount) and view updated projections.	Supports exploration of different planning strategies.
FR-21	The system shall display simulation results as an interactive chart showing projected growth over time.	Makes long-term projections easier to interpret.

4.6 User Account and Financial Data Management (Supporting Module)

Provides secure account access and the data-entry foundation — income, expenses, assets, liabilities, and investments — that all five MSP engines rely on.

ID	Requirement Statement	Rationale
FR-22	The system shall allow users to register and log in using JWT-based authentication.	Establishes secure, unique access to personal financial data.
FR-23	The system shall allow users to create, edit, and delete records of income, expenses, assets and liabilities.	Forms the core data on which all analysis depends.
FR-24	The system shall allow users to record investment holdings including type, amount, and risk level.	Supports investment-aware planning and scoring.
FR-25	The system shall allow users to edit their profile and account settings.	Keeps personal and account information current.
FR-26	The system shall provide password reset functionality via registered email.	Ensures account recovery and continued access.

4.7 Dashboard, Reports, and Goal Tracking (Supporting Module)

Presents consolidated financial information, goal progress, charts, downloadable reports, and the AI Financial Advisor chat interface.

ID	Requirement Statement	Rationale
FR-27	The system shall allow users to create, edit, and track progress of financial goals.	Enables structured, measurable financial planning.
FR-28	The system shall provide a dashboard summarizing net worth, wealth score, goals, and recent AI insights.	Gives users a single, at-a-glance view of their finances.

ID	Requirement Statement	Rationale
FR-29	The system shall render income, expense, and net-worth trends as interactive charts using Chart.js or Recharts.	Makes financial patterns easier to understand visually.
FR-30	The system shall allow users to generate and download a summary financial report.	Supports offline review and record-keeping.
FR-31	The system shall provide a conversational interface for users to ask the AI Financial Advisor questions about their data.	Makes AI-driven insight accessible through natural interaction.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement Statement	Rationale
NFR-01	The system shall respond to standard user queries within 2 seconds under normal load.	Ensures a smooth, responsive user experience.
NFR-02	The system shall return AI-generated explanations within 6 seconds under normal conditions.	Keeps AI-assisted insight interactions practical to use.
NFR-03	The system shall support 1,000 concurrent users without noticeable performance degradation.	Ensures scalability for growing adoption.

5.2 Safety Requirements

ID	Requirement Statement	Rationale
NFR-04	The system shall validate all financial inputs to prevent calculation errors from malformed data.	Protects the accuracy of wealth scores and projections.
NFR-05	The system shall clearly label AI-generated recommendations as informational, not certified financial advice.	Prevents users from over-relying on automated suggestions for major decisions.
NFR-06	The system shall log all critical financial-data operations for audit purposes.	Enables investigation and recovery in case of errors or disputes.

5.3 Security Requirements

ID	Requirement Statement	Rationale
NFR-07	The system shall authenticate all users using JWT tokens with a defined expiry period.	Protects accounts from unauthorized access.

ID	Requirement Statement	Rationale
NFR-08	The system shall encrypt sensitive financial data at rest using AES-256.	Protects stored personal financial data from exposure.
NFR-09	The system shall enforce HTTPS for all client-server communication.	Prevents interception of data during transmission.
NFR-10	The system shall implement role-based access control for user and administrator functions.	Restricts sensitive features to authorized roles only.
NFR-11	The system shall automatically log out inactive sessions after 15 minutes.	Reduces risk of unauthorized access from unattended devices.

5.4 Software Quality Attributes

This section addresses usability, reliability, maintainability, scalability, and explainability.

ID	Requirement Statement	Rationale
NFR-12	The system shall maintain 99.5% uptime, measured monthly.	High availability supports users who rely on the platform for ongoing planning.
NFR-13	The system shall be usable without prior financial training, using plain-language labels and AI explanations.	Broadens accessibility to non-expert users.
NFR-14	The system shall be built with a modular backend architecture separating each analytical engine.	Facilitates independent updates and maintenance of each MSP module.
NFR-15	The system shall support horizontal scaling of backend services as user load grows.	Ensures the platform can grow without a redesign.
NFR-16	The system shall provide an explanation of the key factors behind every AI-generated score, alert, or recommendation.	Builds user trust by making automated outputs transparent rather than a black box.
NFR-17	The system shall recover from a backend service failure without loss of previously saved financial data.	Protects the integrity of user-entered financial records.
NFR-18	The system shall be testable through automated unit and integration test suites.	Reduces regression errors and speeds up quality assurance.
NFR-19	The system shall be portable across major desktop and mobile browsers (Chrome, Edge, Firefox, Safari).	Ensures accessibility across the devices users already own.

5.5 Business Rules

ID	Requirement Statement	Rationale
BR-01	Only authenticated users shall be able to view or modify their own financial data.	Protects the confidentiality of personal financial information.
BR-02	A financial goal shall not be marked complete until its target amount is fully reached.	Maintains accuracy in goal tracking and reporting.
BR-03	Stress test results shall be recalculated whenever the user's income, expenses, or liabilities change materially.	Keeps resilience analysis aligned with the user's current situation.
BR-04	AI-generated recommendations shall always be presented alongside the underlying data used to produce them.	Supports transparency and user trust in automated advice.
BR-05	Goal priority conflicts shall be re-evaluated automatically whenever a new goal is added or an existing goal is edited.	Ensures conflict detection remains current.

6. Other Requirements

ID	Requirement Statement	Rationale
OR-01	The system shall support future integration with bank-account aggregation APIs for automated data import.	Reduces manual data entry and improves data accuracy over time.
OR-02	The system shall comply with applicable data protection regulations (e.g., GDPR-equivalent principles) for personal financial data.	Protects user rights and limits legal exposure.
OR-03	The system shall support scheduled database backup and recovery mechanisms.	Prevents loss of financial data during failures.
OR-04	The system shall allow the AI provider (OpenAI or Gemini) to be swapped via configuration without code changes.	Avoids long-term dependency on a single AI vendor.
OR-05	The system shall support reuse of the scoring, stress-testing, and simulation engines as independent services for future products.	Reduces development cost and effort for future extensions.

Appendix A: Glossary

Term	Definition
Net Worth	Total assets minus total liabilities; the core measure of overall wealth.
Wealth Score	A composite score representing a user's overall financial health.
Stress Testing	Simulating adverse financial events to evaluate resilience.
Reverse Planning	Calculating the contribution needed today to reach a future financial target.
Goal Conflict	A situation where two or more financial goals compete for the same limited funds.
Liquidity	How easily an asset can be converted into cash without significant loss of value.
JWT	JSON Web Token, used for secure, stateless user authentication.
API	Application Programming Interface, used for communication between systems.
AES-256	Advanced Encryption Standard with a 256-bit key, used for secure data encryption.
HTTPS	Hypertext Transfer Protocol Secure; ensures encrypted communication over the web.

Appendix B: Analysis Models

The following analysis models support the requirements defined in this document, illustrating system behavior, data flow, structure, and relationships.

B.1 Use Case Diagram

Actor: the individual User, who interacts with all ten core use cases. Six of these use cases in turn depend on the AI Advisory Sub-System to generate explanations and recommendations.

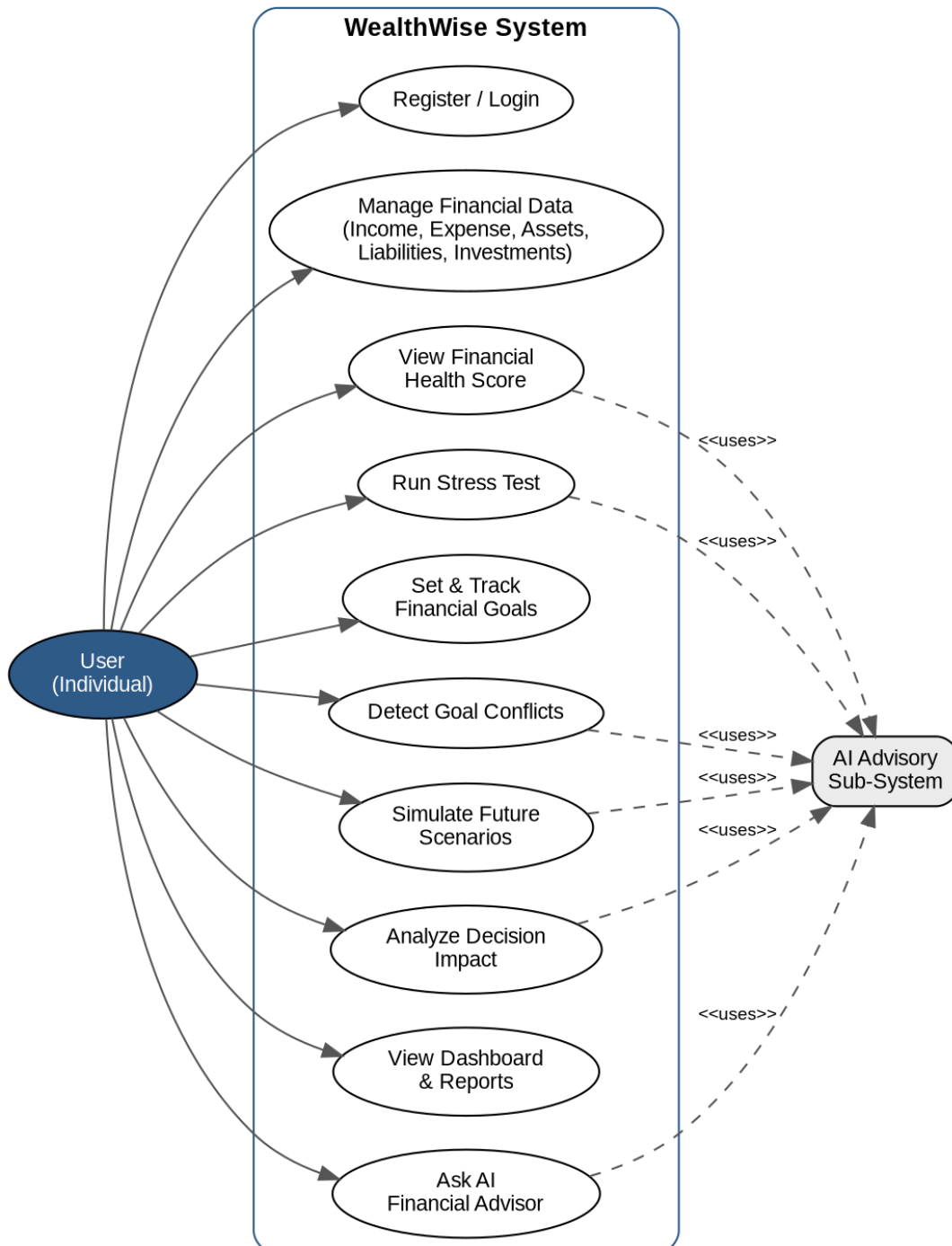
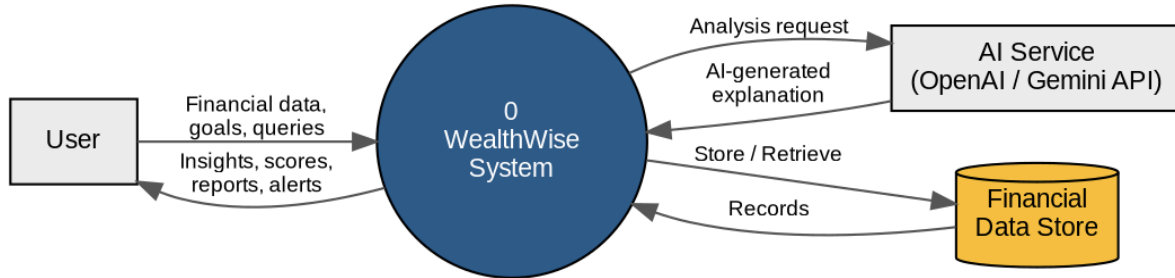


Figure B.1 – Use Case Diagram: User Interactions with WealthWise

B.2 Data Flow Diagram (Level 0)

The context-level DFD shows WealthWise as a single process interacting with the User and the external AI service, backed by a financial data store.

Figure B.2 – Data Flow Diagram, Level 0 (Context Diagram)



Appendix C: To Be Determined List

TBD ID	Description
TBD-01	Final selection of the AI provider (OpenAI API vs. Gemini API) for production use.
TBD-02	Decision on whether bank-account aggregation will be added in a future release.
TBD-03	Choice of cloud hosting provider (AWS, Azure, or GCP).
TBD-04	Finalization of supported currencies for multi-currency users.
TBD-05	Decision on a dedicated mobile app (React Native or Flutter) beyond the responsive web app.